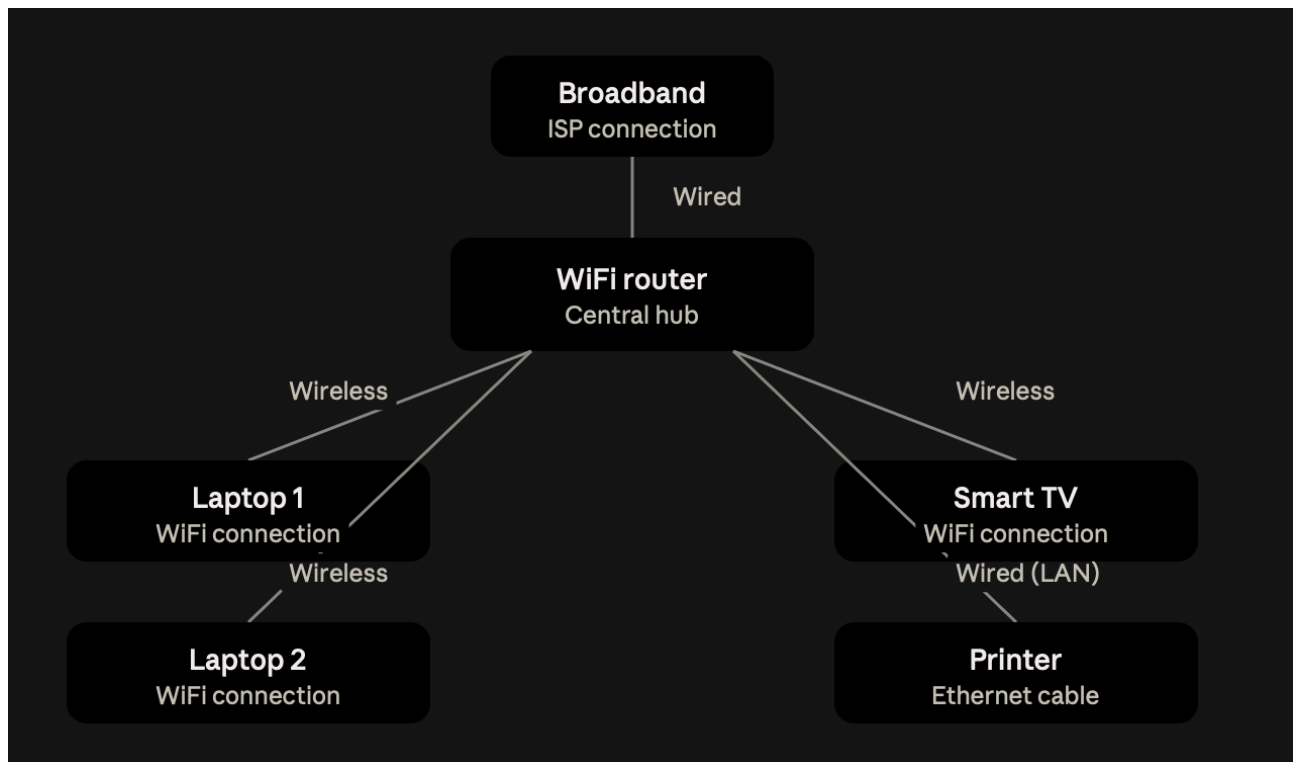


Module 2: Network + Network Security , Maintenance and Troubleshooting

Build a SOHO network

Task 1:



Task 2:

Component	Suggested Model	Notes
WiFi router	TP-Link Archer C6 (AC1200, dual-band, ~₹2,500) or TP-Link Archer AX21 (WiFi 6, slightly pricier)	Dual-band is enough for 5 devices; WiFi 6 future-proofs if you add more
Ethernet cable	Any Cat6 cable, 3–5 meters	Cat6 is the current standard, backward-compatible and cheap

Component	Suggested Model	Notes
Printer	Canon PIXMA TS4320 (wireless, all-in-one) or HP DeskJet 2855e	Both support WiFi directly, but since your diagram uses wired for the printer, either also has an Ethernet/USB port for a stable wired link
Laptops	Any laptop with built-in WiFi	No extra hardware needed — built-in WiFi adapter is standard
Smart TV	Any smart TV with built-in WiFi	Same — no extra hardware needed

Task 3:

1. Connect to the router's admin panel

- Plug a laptop into the router via Ethernet (or connect to its default WiFi), then open a browser and go to the router's default gateway address, usually 192.168.0.1 or 192.168.1.1 (check the sticker on the router itself for the exact address and default login).

2. Log in with default credentials

- Enter the default username/password from the router's sticker (often admin/admin) to access the settings dashboard.

3. Change the default admin password

- Go to System Tools or Administration settings and immediately change the default admin password to something strong. This is critical — default passwords are publicly known and the first thing attackers try.

4. Set a unique SSID

- Under Wireless Settings, change the network name (SSID) from the factory default (like "TP-Link_1234") to something unique that doesn't reveal your address, name, or router brand — e.g. avoid "John's House WiFi".

5. Enable WPA2 (or WPA3) security

- In the same Wireless Settings section, set the security mode to WPA2-PSK (or WPA3 if your router and devices support it) — never leave it open or use the outdated WEP standard, which is trivially crackable.

6. Set a strong WiFi password

- Choose a passphrase of at least 12 characters mixing letters, numbers, and symbols — this is separate from the admin password and is what devices use to join the network.

- **7. Disable WPS if not needed**

- WiFi Protected Setup (the push-button pairing feature) has known vulnerabilities. Turn it off under Wireless Settings unless you specifically need it for a device that only supports WPS pairing.

- **8. Save and reboot the router**

- Apply all changes and reboot the router. Reconnect your devices using the new SSID and password to confirm everything works.

Task 4:

Device	Assignment type	MAC address	IP address
Laptop 1	DHCP	AA:BB:CC:11:22:33	192.168.1.101
Smart TV	Static	AA:BB:CC:77:88:99	192.168.1.50
Printer	Static	AA:BB:CC:AA:BB:CC	192.168.1.51

Task 5:

WiFi connection issue on the laptop

The laptop may have dropped off the WiFi network entirely, or connected to the wrong/weak-signal network.

Solution:

- Check the WiFi icon — confirm it shows connected to your actual SSID
- Try forgetting the network and reconnecting with the password
- Move closer to the router to rule out a weak signal

IP address / DHCP problem

The laptop may have failed to get a valid IP address from the router, or has an IP conflict with another device.

Solution:

- Run ipconfig (Windows) or ifconfig (Mac/Linux) to check if it has a valid 192.168.x.x
- If it shows a 169.254.x.x address, that means DHCP failed — try ipconfig / release
- Restart the router to refresh its DHCP pool

Router or ISP-level outage

The problem may not be the laptop at all — the router or the broadband line itself could be down.

Solution:

- Check if other devices (phone, TV) also can't reach the internet
- Look at the router's status lights — a red or blinking WAN light usually signals an ISP
- Power-cycle the router and modem (unplug for 30 seconds, plug back in)

DNS resolution failure

The laptop may have network connectivity but can't translate website names into IP addresses.

Solution:

- Try pinging a raw IP like 8.8.8.8 — if that works but google.com doesn't, it's a DNS issue
- Flush the DNS cache (ipconfig /flushdns on Windows)
- Manually set DNS to 8.8.8.8 (Google) or 1.1.1.1 (Cloudflare) in network settings

